Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

6.5930 Project 12: Evaluate a New Technology

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Emerging technology study: Lightmatter Passage photonic interconnect

The One-Slide Thesis

42.4×

Best modeled inference latency speedup. Gemma 4 26B mixture-of-experts on a 9216-chip pod.

0 of 24

Local mapping signatures that change under the photonic swap on the 64-chip torus.

5 of 16

Distributed schedule choices that change once multicast and pooled remote memory become cheap.

The claim

Photonic interconnect is not just a lower-pJ/bit wire. Where it really earns its keep is enabling a different distributed schedule above the local mapping: multicast cache fill, remote pooled-HBM streaming, and a flatter, switched fabric in place of the long electrical hops.

Why This Belongs in a TPU/LLM Conversation

- TPU7x/Ironwood is publicly documented at 9216 chips per pod, 192 GiB of HBM per chip, and 1200 GB/s of bidirectional inter-chip bandwidth per chip.
- Decode caches, expert tokens, gradients, and shared weights are all moved across chips during normal training and inference.
- Production software already treats the pod as a sharded device mesh: GSPMD-style annotations choose where data lives and the compiler inserts the matching collectives.

The question we asked

If a future TPU pod were wired with Lightmatter-style photonics, would the mapper just see cheaper bits, or would the optimal placement and schedule above the mapper actually change?

The Experiment Spine

Six experiments, one question each, in the order they were run

1. 16-node AccelForge feedback loop

Q: Does a logical-memory network level change energy and latency?

A: Yes: optical multicast reduces cluster energy by ~72% and latency by ~98%.

2. Distance-aware link model

Q: Is pJ/bit constant with scale?

A: No: electrical/chiplet links grow with distance; photonics is conversion dominated.

3. TPU cube mapping and multicast

Q: Is the gain just bandwidth?

A: No: shared-state multicast collapses N unicasts into one broadcast.

4. Mapping vs distributed schedule

Q: Can the AccelForge mapping stay stable while compiler choices change?

A: Yes: 0/24 inner signatures change, but 5/16 distributed choices change.

5. Strong/weak scaling and OCS/OPS

Q: What improves with larger systems and switching semantics?

A: Photonic multicast scales flatter; OPS mainly helps fine-grained phases.

6. 9216-chip Gemma/Gemini proxy

Q: Does the stack matter end to end?

A: Inference improves far more than training because training becomes compute-bound.

What “Mapping” Means in This Study

1. **Workload:** a parameterized GEMM einsum that stands in for a particular LLM stage.
2. **Architecture:** the per-chip memory hierarchy plus a logical cluster-memory level above it.
3. **Inner mapping:** the loop order, tile shape, and tensor residency that the mapper picks locally.
4. **Distributed schedule:** the compiler-visible decision about whether traffic is point-to-point, fetched from a remote pool, or multicast from a shared source.

Why the split matters

The inner mapping can stay identical while the distributed schedule above it changes, and that is exactly what we see in our experiments. Reporting them together would have hidden the actual mechanism.

Network Cost Folded Back Into Memory

$$E_{\text{net}} = B\big(e_c + h\,\ell(N)\,e_{tx}\big)$$

$$T_{\text{net}} = \frac{B}{BW/C(N)} + h\,t_r + h\,\ell(N)\,t_p + t_s$$

- B: bits transferred, taken from mapper action counts.
- h: average hop count for the topology.
- $\ell(N)$: average physical path distance.
- $C(N)$: contention factor under N active nodes.
- t_s: optical-switching reconfiguration term.

Iterative loop

Mapper → logical-memory traffic → topology cost → updated logical-memory energy and latency → mapper rerun, until the per-bit numbers stabilize.

The multicast rule

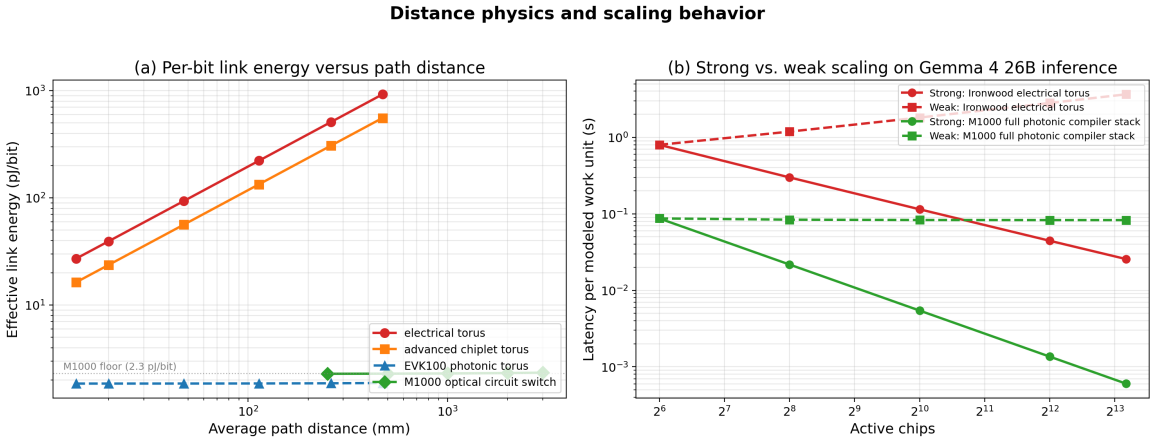
Electrical and chiplet baselines pay every receiver. Multicast-capable photonics pays the unique payload exactly once.

Assumptions Kept Visible

Item	Value	Role
Ironwood pod	9216 chips; 192 GiB HBM/chip; 1200 GB/s ICI/chip	Full-system baseline
M1000 photonic	57.3 Tb/s one-way, 2.3 pJ/bit including laser	Main photonic fabric
EVK100 photonic	3.2 Tb/s class, 1.9–3.2 pJ/bit	Conservative and optimistic sensitivity points
Electrical link	0.2 pJ/bit conversion + 1.96 pJ/bit/mm transmission	Distance-sensitive baseline
Chiplet baseline	3.5 Tb/s, 6 pJ/bit/hop	Strong non-optical comparison
Compiler layer	SPMD collectives, remote pooled HBM, replicated cache, photonic multicast	GSPMD-style schedule choices

The compute side, the per-chip memory hierarchy, and the HBM cost are held fixed across every comparison. Only interconnect, topology, multicast, and the distributed schedule change.

Result 1: Distance Reshapes the Scaling Law



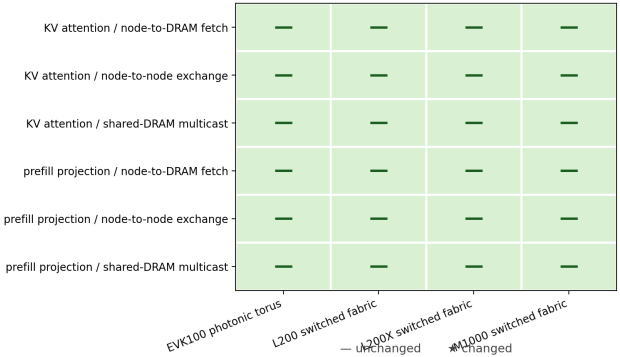
Reading Result 1

- Electrical and chiplet links are good at very short reach but accumulate transmission energy with distance.
- Photonic links pay their conversion energy once and stay nearly flat, so the curves cross over before pod scale.
- Strong scaling and weak scaling are kept separate to avoid conflating “faster on fixed work” with “more aggregate throughput.”
- This is also the reason a constant-pJ/bit comparison would have been misleading: it would have hidden which side of the crossover each topology lives on.

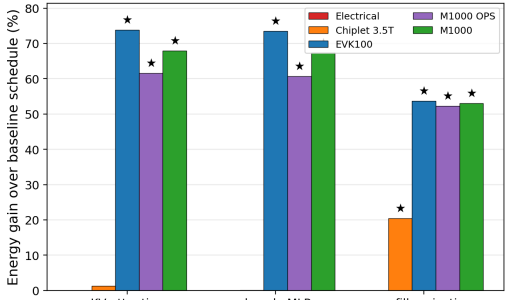
Result 2: The Local Mapping Does Not Move

Local mapping is stable; the distributed schedule is what changes

(a) Inner mapping signature relative to electrical baseline



(b) Distributed-schedule energy gain (★ = strategy changed)

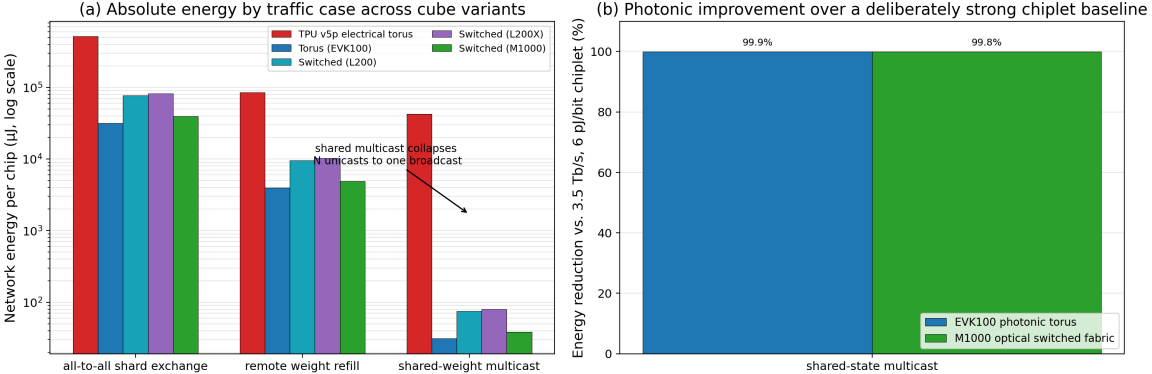


Reading Result 2

- Across all photonic torus and switched-fabric variants on the 64-chip cube, the local mapping signature is the same as the electrical baseline. Every cell on the left is unchanged.
- One layer up, the picture is the opposite. Photonic technologies prefer remote pooled-HBM streaming for decode and KV attention, and they prefer shared-DRAM multicast for prefill projection.
- The electrical and chiplet baselines see almost no benefit from those alternative schedules, because the corresponding traffic is too expensive on those links.
- This is the cleanest evidence we have that the photonic gain is mostly above the mapper, in compiler-visible territory.

Result 3: Multicast and the Strong Chiplet Baseline

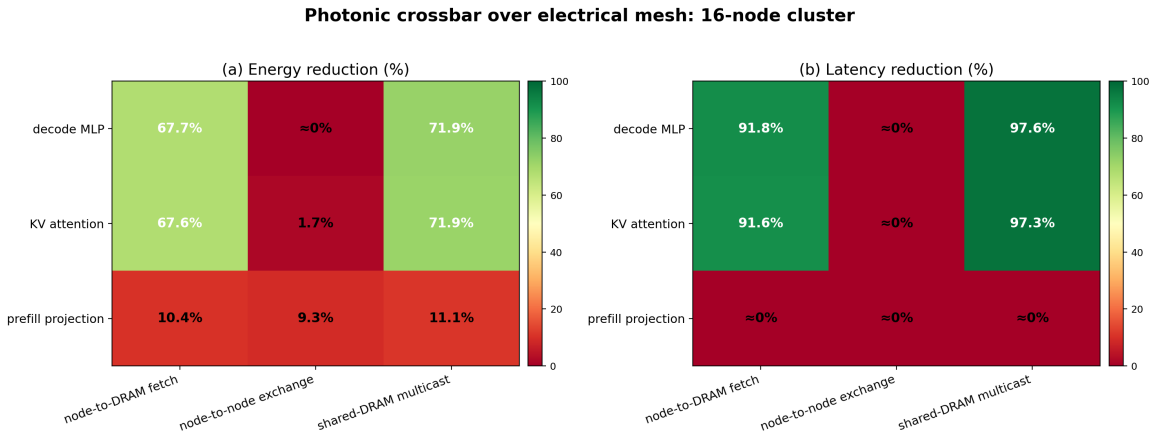
Multicast and topology semantics, not raw bandwidth, are decisive



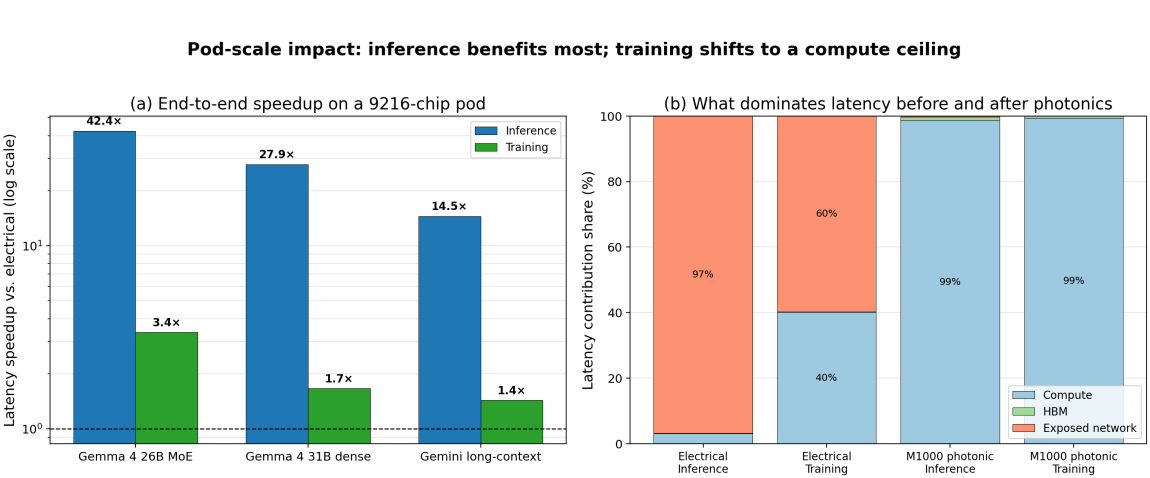
Reading Result 3

- The chiplet baseline is deliberately not weak: 3.5 Tb/s and 6 pJ/bit/hop, similar to advanced packaging targets.
- The shard-exchange and weight-refill cases are roughly an order of magnitude below the electrical torus across photonic variants. The shared-multicast case is closer to four orders of magnitude below it.
- The reason is structural rather than bandwidth-driven: shared multicast collapses N unicast deliveries into one logical broadcast. That is a topology-semantics effect, not a wire effect.
- Workloads that cannot exercise that pattern see far smaller gains. We keep both kinds of cases in view rather than reporting only the favorable ones.

Result 4: 16-Node Validation Heatmap



Result 5: End-to-End Pod Impact

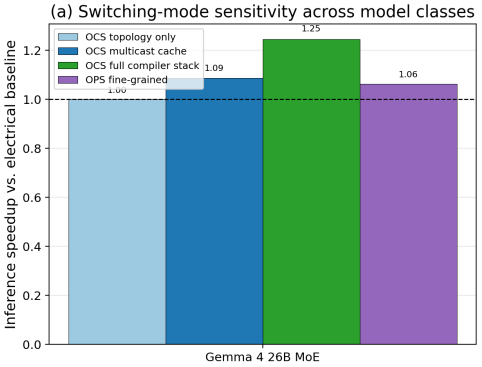


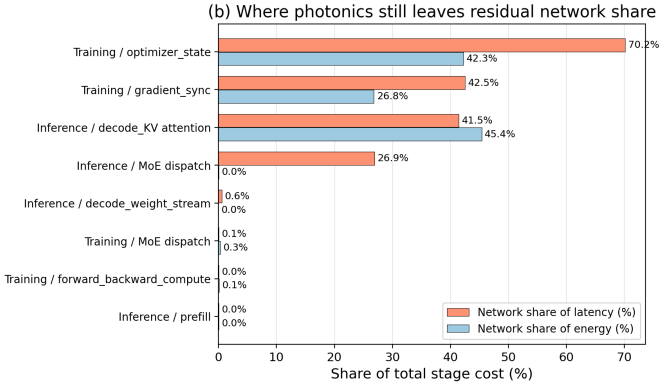
Reading Result 5

- Inference latency on the Gemma 4 26B mixture-of-experts proxy improves by roughly $42\times$. The long-context Gemini proxy improves by roughly $15\times$.
- Training improvements are far smaller—about $3.4\times$ and $1.4\times$ respectively—because once communication shrinks, what is left is matrix-multiply compute.
- The right panel makes the regime change concrete. Electrical inference is overwhelmingly network-bound. Photonic inference is overwhelmingly compute-bound. That is the textbook signature of a successful interconnect intervention.
- Training was already more compute-balanced, so its remaining headroom is smaller.

Result 6: Switching Mode and Residual Network

Switching mode and residual bottlenecks after photonics

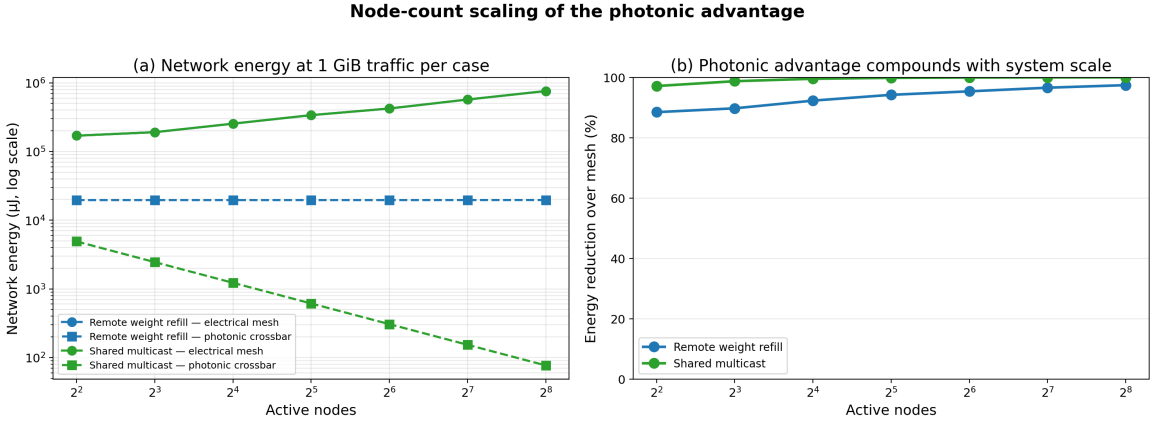




Reading Result 6

- Topology alone, with no compiler change, brings the system to roughly parity with the electrical baseline. Layering multicast cache fill on top adds a small but measurable lift.
- Running the full compiler stack with multicast and remote pooled HBM is what moves the inference speedup into the $1.06\times$ to $1.25\times$ band relative to a topology-only switch baseline.
- Optical packet switching is included as a sensitivity point. It is not where most of the gain comes from—multicast and topology are.
- After photonics, the residual network share is concentrated in training optimizer-state and gradient-sync stages, which is where future work should focus.

Node-Count Scaling



Headline Numbers

Layer of the study	Result	Interpretation
16-node cluster	71.95 % energy, 97.57 % latency reduction	Logical-memory feedback loop visibly changes system metrics
64-chip cube network	99.93 % energy reduction	Multicast collapses repeated traffic
64-chip cube mapping	$0/24$ inner mappings change	Local dataflow stays put
Distributed schedule	$5/16$ choices change	Compiler-visible mapping is where the change lives
Compiler strategy	73.86 % best energy gain	Remote pooled-HBM and multicast cache fill can be optimal
Pod inference (Gemma 4 MoE)	$42.4\times$ speedup, 94.05 % energy reduction	Decode, cache, and expert routing benefit most
Pod training (Gemma 4 MoE)	$3.4\times$ speedup	Communication shrinks, then compute dominates

What Worked, What Did Not

Where photonics earned its place

- Distance-aware physics at pod scale.
- Hardware multicast for shared state.
- Remote pooled-HBM streaming and multicast cache fill.
- Decode-heavy and mixture-of-experts inference.

Where it did not

- Local GEMM mappings did not move in the tested cases.
- Pure point-to-point unicast traffic gained the least.
- Training became compute-bound after collectives shrank.
- Detailed packet queueing, area and cost, thermal drift, and exact compiler code generation are out of scope.

Reproducibility and Validation

- All quantitative results in these slides come from the same end-to-end study script, executed end-to-end, with both raw per-experiment data and a compact submission package committed to the repository.
- Every figure in this deck and in the report is regenerated from the same data tables. There are no hand-edited plots.
- A regression suite enforces the central physics (electrical-versus-photonic distance crossover), the multicast scaling law, the strong/weak scaling separation, the inner-mapping invariance claim, and the optical-switching sensitivity, so a future tweak to the catalog cannot silently invalidate the headline claims.

References


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Takeaway

Final answer

Photonic interconnect is most valuable when the system above it can expose multicast, pooled memory, and flatter topologies to the compiler. The local accelerator mapping can stay where it is. The pod-level distributed schedule is what moves, and that movement is where the inference speedup lives.

Questions